ST MARY'S DSG, KLOOF

GRADE: 10

2 June 2026

PHYSICAL SCIENCE CLASS TEST
Magnetism and Electrostatics

TIME: 40 MINS

ASSESSOR: K. Bolton

MODERATORS: G. Main & T. Kirsten

TOTAL: 42 40 *

NAME: MEMO

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 8 pages. A data sheet is provided.
2. Answer on the question paper in the spaces.
3. Write your name on your question paper.

$Q=nq_e$

Charge on a single electron is $q_e=1,6\times10^{-19}$ C

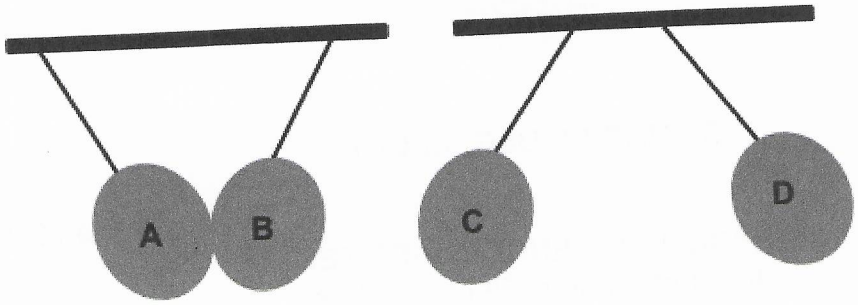
QUESTION 1 – MULTIPLE CHOICE

For the next three questions you must make a choice out of the answers A - D given. **On the grid provided, make a cross over the letter you think is most correct, e.g.**

1.4	A	~~B~~	C	D

1.1	A	~~B~~	C	D
1.2	A	B	~~C~~	D
1.3	A	~~B~~	C	D

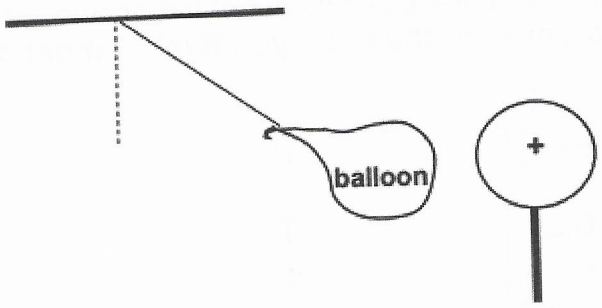
1.1 Four identical balloons, each carrying a charge, are suspended from a ceiling, as shown in the diagram below.



Balloon B is negatively charged. Which combination is CORRECT regarding the charges on the balloons?

	SIGN OF CHARGE A	SIGN OF CHARGE C	SIGN OF CHARGE C
A	-	+	-
B	+	+	+
C	-	-	-
D	+	+	-

1.2 A balloon is brought closer to a positively charged sphere as shown in the diagram below. The balloon is attracted to the sphere. What is the type of charge on the balloon?

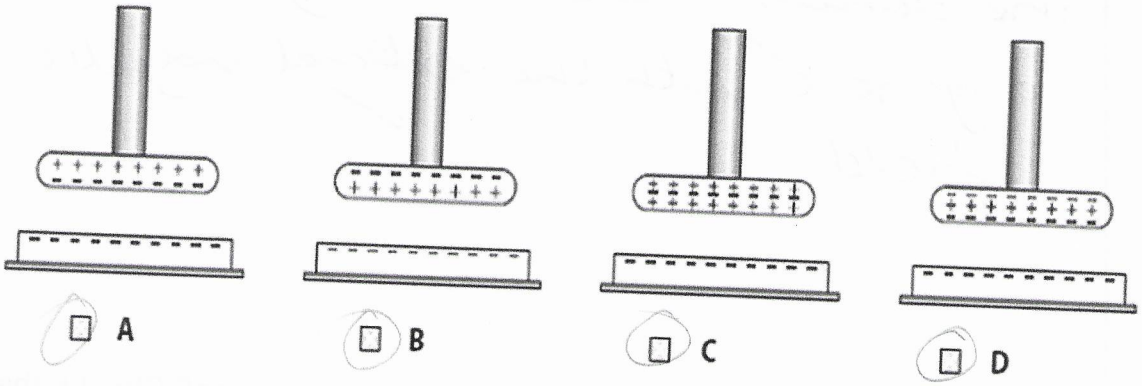


A positive
B positive or neutral
C negative or neutral
D negative

1.3 The diagram below shows a plastic block and a metal disc with an insulating handle. The top surface of the plastic block has a negative charge. The metal disc has no overall electric charge.



A student uses the insulating handle to hold the metal disc above the plastic block. Which of these diagrams shows how the charge is distributed on the metal disc?



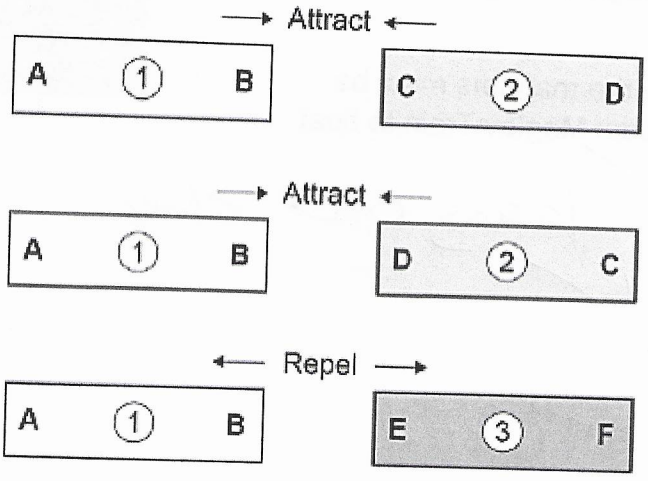
can you blank these out?

[3x2=6]

QUESTION 2

2.1 Each of the three metal bars in the diagram below is either a bar magnet or a piece of unmagnetised iron.

The forces that act between the bars when different ends are placed close together are shown by the arrows.



2.1.1 Iron is a soft magnetic material. Explain what is meant by this term. (2)

Can be easily magnetised

2.1.2 Use the domain theory to explain why a piece of unmagnetized iron is attracted to a bar magnet. (3)

The domains in the iron swing into alignment with the external magnetic field

2.1.3 Which one of the metal bars is a piece of unmagnetised iron? CIRCLE the correct answer. (2)

Bar 1 **Bar 2** **Bar 3**

2.2 The Maglev train floats over its track using an electromagnet. Maglev trains have managed speeds in excess of 600 km/h. A model of the Maglev train uses two permanent magnets to get the model to float over the track.



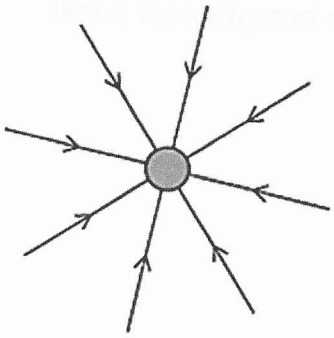
Describe how and why the magnets must be arranged to get the model Maglev Train to float. (2)

Like poles facing each other to repel

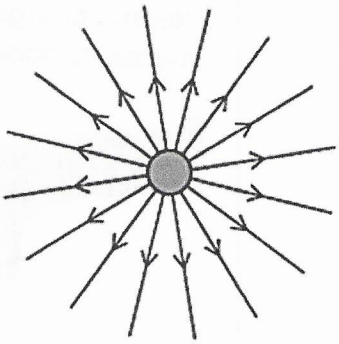
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3.2 Two small objects, P and Q, are each given an electric charge. The diagram below represents the electric fields around the objects, P and Q.



object P



object Q

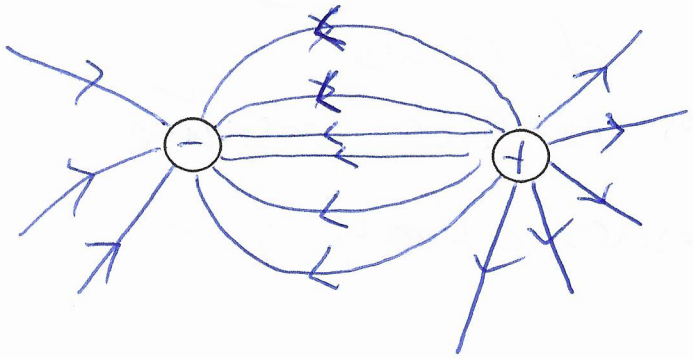
3.2.1 Use information from the diagram to give TWO differences between the charge on P and the charge on Q. (2)

- P ⊖ + Q ⊕ ✓
- Q more charge than P ✓

3.2.2 Define the term *electric field*. (2)

Region of space in which an electric charge experiences a force

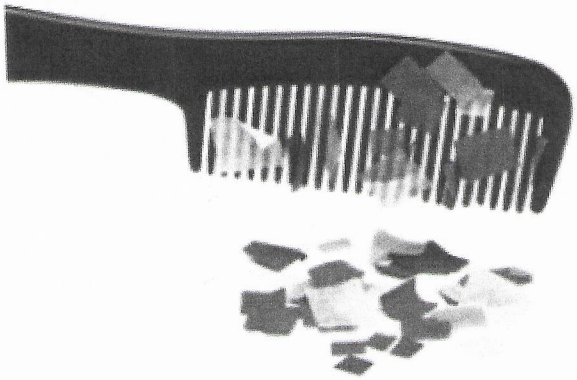
3.2.3 Object P and object Q are held near to each other so that their electric fields interact with each other. In the space below draw the electric field between the two charges. (3)



Shape ✓
arrows ✓
more lines ⊕ ✓

QUESTION 3

3.1 A student rubs a PVC comb with a dry cloth to give the comb a negative electric charge. The the charged PVC comb is then brought near small pieces of paper which it is picking. up



(Source © GIPhotoStock/SCIENCE PHOTO LIBRARY)

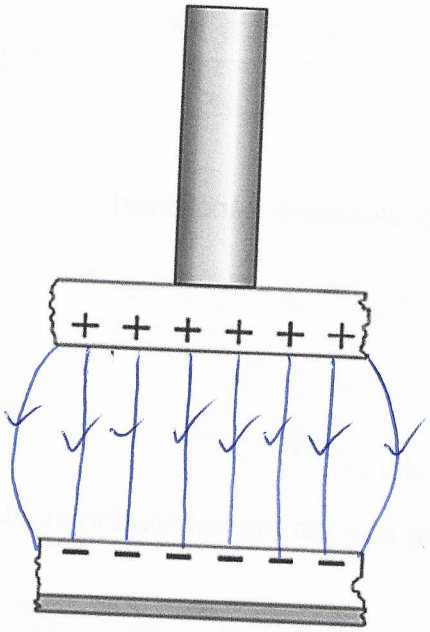
3.1.1 Explain how rubbing the comb with a dry cloth gives the comb a negative electric charge. (3)

The comb gains electrons from the cloth

3.1.2 Explain how the negatively charged plastic comb picks up the small pieces of paper. (3)

The electrons in the pieces of paper are repelled away from the surface due to negative charge of comb. Leaves surface positive ∴ opposites attract

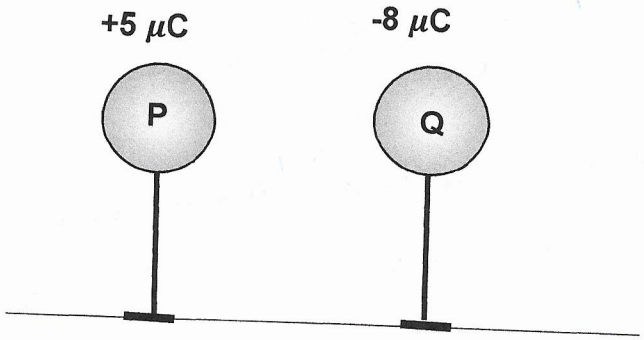
3.3 The diagram below shows the charges on part of the metal disc and a plastic block.



arrows ✓
equidistant ✓

On the diagram above, draw lines to show the shape and direction of the electric field between the metal disc and the plastic block. (2)

3.4 Two spheres P and Q, placed on insulated stands, carry charges of +5 μC and -8 μC respectively.



The charges are brought into contact and are then separated.

3.4.1 Are electrons transferred from P to Q or from Q to P during contact? Give a reason for the answer. (2)

Q to P ✓

Q has an xcess of electrons ✓

3.4.2 Calculate the new charge on P and Q after contact. (2)

$$Q_{new} = \frac{Q_1 + Q_2}{2} = \frac{+5 + (-8)}{2} \checkmark m = -1,5 \checkmark \mu C$$

3.4.3 Calculate the number of electrons transferred. (3)

$$n = \frac{\Delta Q}{q_e} = \frac{-1,5 \times 10^{-6} - 5 \times 10^{-6}}{-1,6 \times 10^{-19}} \checkmark conv \checkmark m \quad OR \quad \frac{-1,5 \times 10^{-6} - (-8 \times 10^{-6})}{1,6 \times 10^{-19}}$$

$$= 4,06 \times 10^{13} \checkmark$$

3.4.4 Name and state the law used in the calculation in QUESTION 3.4.2. (3)

Law of conservation of charge ✓

No. of e's lost = no. of e's gained ✓✓

Charge cannot be created or destroyed

check size of font

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